GOLD IN BIOLOGICAL ENVIRONMENTS: A MARTINI MODEL

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ABSTRACT

Gold is used in a variety of biomedical and drug delivery applications, yet it is difficult to reliably model gold interactions with biological materials in the Martini force field. In a previous study1, we showed that aggregation of existing Martini nanoparticle models used to represent gold nanoparticles depends on the choice of gold-core parameters. However, a Martini gold bead model that simultaneously accounts for gold–gold and gold–biomolecule surface affinities has not been developed. Here we develop a Martini coarse-grained gold bead designed as a counterpart to the INTERFACE force field. The model was designed to preserve the structural, energetic, and interfacial properties of gold. To parameterize the gold model, we match several targets to their atomistic or experimental counterparts, including gold-gold self-interaction energies, gold-biomolecule adsorption free energies. We then performed unbiased nanoparticle dimer simulations initialized without aggregation. These simulations produced a dimer with interparticle spacings qualitatively consistent with those observed in experimental cryo-EM images. We will further test the model through quantitative comparisons of simulated and experimental interparticle-distance distributions. We also plan to formalize the parameterization framework for extension to other metals.